ONE-PAGE BRIEF

Post-quantum cryptography review: secp256k1 @ 1b4181f93f7a

Client: Bitcoin Core (libsecp256k1) — public repository (passive review) · Source: github.com/bitcoin-core/secp256k1, commit `1b4181f93f7a` · Reviewed: 2026-11-04 19:25 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 0 of 4

no post-quantum cryptography

FINDINGS

0 Critical **0** High **4** Medium

0 Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

Medium	QSI-2026-12813	**Quantum-vulnerable ECDSA: transaction and message signing (10 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12815	**Quantum-vulnerable Schnorr (BIP-340): transaction and message signing (3 file(s))** confirmed by the analyst What to do: Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum signature schemes as chains...
Medium	QSI-2026-12812	**Quantum-vulnerable ECDSA: role not determined (13 file(s))** confirmed by the analyst What to do: Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205), in a hybrid/composite form...
Medium	QSI-2026-12814	**Quantum-vulnerable ECDH: role not determined (4 file(s))** confirmed by the analyst What to do: Classify each usage, record data lifetimes, and plan migration to Hybrid key establishment with ML-KEM (FIPS 203), e.g. X25519MLKEM768 in...

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium transaction and message signing	ECDSA, Schnorr (BIP-340)	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	ECDSA, ECDH	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Analyst review: 4 confirmed, 0 not applicable, 0 dismissed as false positives. (QSI review, 2026-10-29)

Static review of source code and manifests; files analysed: 90. No requests were sent to the owner's systems.

Full report, content SHA-256: `df6c0ce6489e5158bf5c2d7d7168387c8cea5fc39bdd0e3f82e73231addc5521`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.